

Ilyana Anderson

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Education

The University of Texas at Austin

M.S. in Computer Science, GPA: 4.0

Coursework: Deep Learning, Advanced Operating Systems, Natural Language Processing, Android Programming, Online Learning and Optimization, Advanced Linear Algebra for Computing, Machine Learning, Case Studies in Machine Learning, Optimization, Reinforcement Learning.

2022 – 2024

Austin, TX

University of Houston

Coursework for the Ph.D. program in Chemistry was completed without obtaining the degree, GPA: 3.835

2015 – 2017

Houston, TX

Lomonosov Moscow State University

B.S. and M.S. in Geology, With Honors

2006 – 2012

Moscow

Technical Skills

Programming Languages: Python, C, SQL

Frameworks and Libraries: PyTorch, NumPy, Pandas, Matplotlib, Fastai, Hugging Face Transformers, Scikit-Learn

Version Control: Git

Project Experience

Prediction of Nucleotide Reactivity at Each Position in RNA Molecules: I used a customized transformer architecture to predict two types of reactivity at each position of RNA molecules as part of the Stanford Ribonanza RNA Folding competition on Kaggle. I experimented with rotary and sinusoidal embeddings for encoding positional information, as well as with feature design. My solution placed in the top 8% of the contest submissions.

Exploration of Methods to Remove Bias From LLMs Fine-Tuned on Natural Language Inference Task: To reduce bias, I developed two novel approaches and implemented the DRIFT method reported in the literature. Most of the enhanced models demonstrated improved performance on the HANS dataset compared to the baseline, suggesting a potential reduction in bias.

Development of a Neural-Network-Based Agent to Play SuperTuxKart Ice Hockey: I implemented imitation learning and DAGGER algorithms within a team project. The team selected my agent design as the basis for further experimentation. This agent placed second in a class-wide competition.

Work Experience

Lamar University

Enrollment Management Analyst

- Submitted enrollment data to National Student Clearinghouse.

2026 – Present

The University of Texas at Austin, Computer and Data Science Online

Learning Facilitator for Natural Language Processing and Advances in Deep Learning

- Responded to students' questions on the discussion board, explained course concepts and assignments.
- Conducted office hours, helped with debugging and code-related issues.
- Collaborated with the professor to improve course content, provided feedback on instructional materials, helped with developing exam questions.

2024 – 2026

Remote

Magoosh

Remote Test Prep Expert

- Provided academic support across multiple standardized tests including MCAT, LSAT, GRE, and GMAT.
- Offered tailored problem-solving strategies and facilitated effective test preparation.

2019 – 2021

Remote

University of Houston, Chemistry Department

Teaching Assistant (TA)

- Led chemistry labs through brief instructional lectures, hands-on supervision, and student support.

2015 – 2017

Houston, TX

Certificates (MOOCs)

Probability - The Science of Uncertainty and Data (MITx)

CS50's Web Programming with Python and JavaScript (HarvardX)